

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

**COURSE NAME: Cloud Computing and
Big Data Analytics**

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)

Question Count: 3

Total Marks: 30

Code of Conduct

I S.Nitheesh (192524329) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Nitheesh

Case Study 1: Smart Retail Inventory Management System

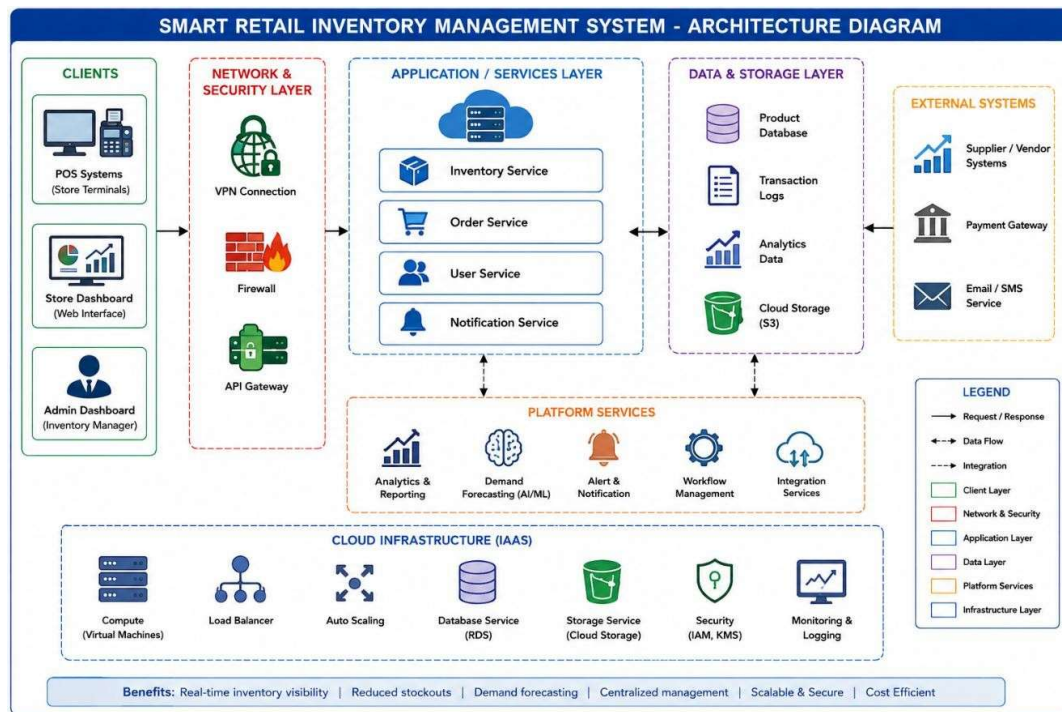
Overview

A retail chain uses a cloud-based inventory management system to monitor stock, sales, and customer demand across multiple stores. Store Point of Sale (POS) systems and dashboards communicate with cloud services using APIs.

Components

- **Clients:** POS systems, Store Dashboards
- **Web APIs:** Enable communication between clients and cloud services
- **Cloud Services:** Inventory management, demand forecasting, analytics
- **Cloud Storage:** Product database, transaction logs, analytics datasets
- **Platform Services:** Data analytics, AI-based demand prediction
- **Infrastructure:** Virtual servers, networking, load balancers
- **Security:** VPN, Firewalls, API Gateway

Architecture Diagram



Cloud Service Models

- IaaS – Virtual Machines, Networking
- PaaS – Analytics Platform

- SaaS – Inventory Dashboard

Benefits

- Real-time inventory updates
- Automatic stock replenishment
- Demand forecasting
- High scalability
- Centralized management
- Secure communication

Security Features

- VPN connectivity
- Firewall protection
- API Gateway
- Authentication
- Data encryption

Conclusion

Cloud computing improves inventory accuracy, reduces operational costs, and enables intelligent business decisions.

Case Study 2: Cloud-Based Document Collaboration System

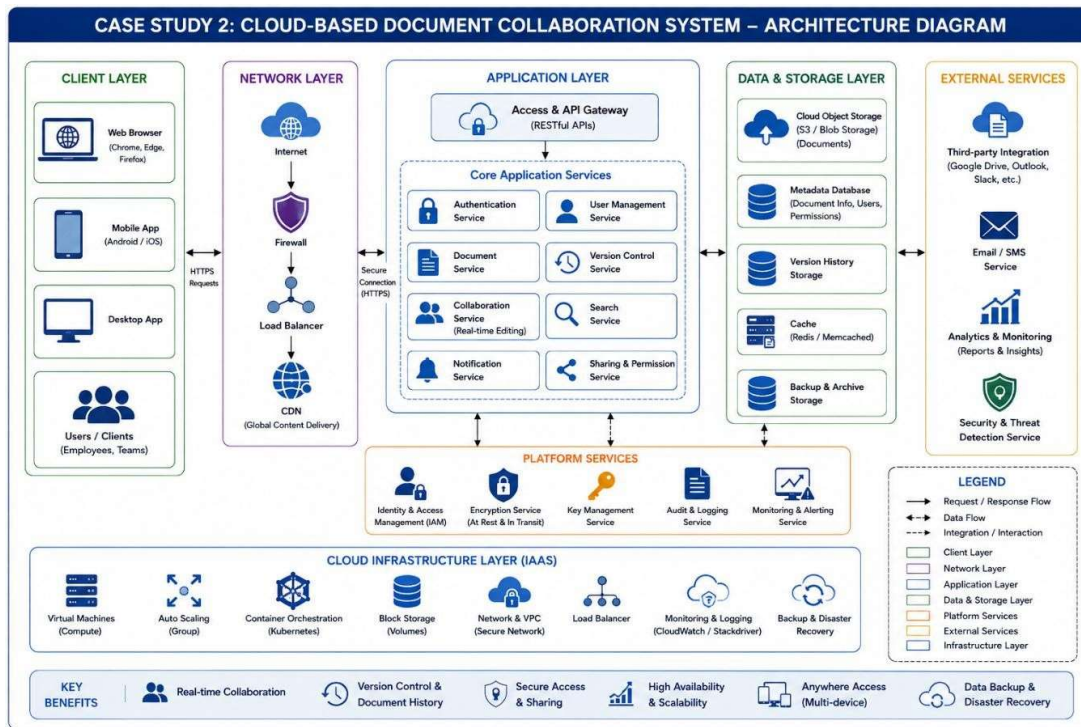
Overview

An organization deploys a cloud-based document editing platform similar to Google Docs, allowing multiple users to edit documents simultaneously from different devices.

Components

- **Clients:** Web browsers, Mobile applications
- **Web APIs:** Document synchronization
- **Cloud Storage:** Distributed document storage
- **Version Control:** Maintains document history
- **Network Infrastructure:** Global CDN and Load Balancers
- **Security:** Encryption, Identity Management, Access Control

Architecture Diagram



Cloud Service Models

- SaaS – Online Document Editor
- PaaS – Collaboration Platform
- IaaS – Cloud Infrastructure

Features

- Real-time editing
- Auto-save
- Version history
- Multi-user collaboration
- File sharing
- Cross-device synchronization

Security Features

- HTTPS encryption
- Identity and Access Management (IAM)
- Role-based permissions

- Secure authentication
- Data encryption at rest

Benefits

- Global accessibility
- High availability
- Easy collaboration
- Automatic backup
- Disaster recovery

Conclusion

Cloud collaboration systems improve productivity by allowing secure, real-time document sharing across the globe.

Case Study 3: Online Food Ordering Platform

Overview

A startup develops a cloud-based food ordering platform that allows customers to browse menus, place orders, and make payments through web and mobile applications.

Components

- **Clients:** Web Browser, Mobile App
- **REST APIs:** Communication between frontend and backend
- **Backend Services:** Order processing, payment handling
- **Cloud Storage:** Menu images, invoices, customer records
- **Infrastructure:** Virtual Machines and Kubernetes
- **Security:** Authentication Tokens, HTTPS, RBAC

Cloud Service Models

- SaaS – Food Ordering Application
- PaaS – Kubernetes Platform
- IaaS – Virtual Machines and Networking

Features

- Online ordering
- Restaurant management

- Order tracking
- Payment integration
- Notification service
- Customer profile management

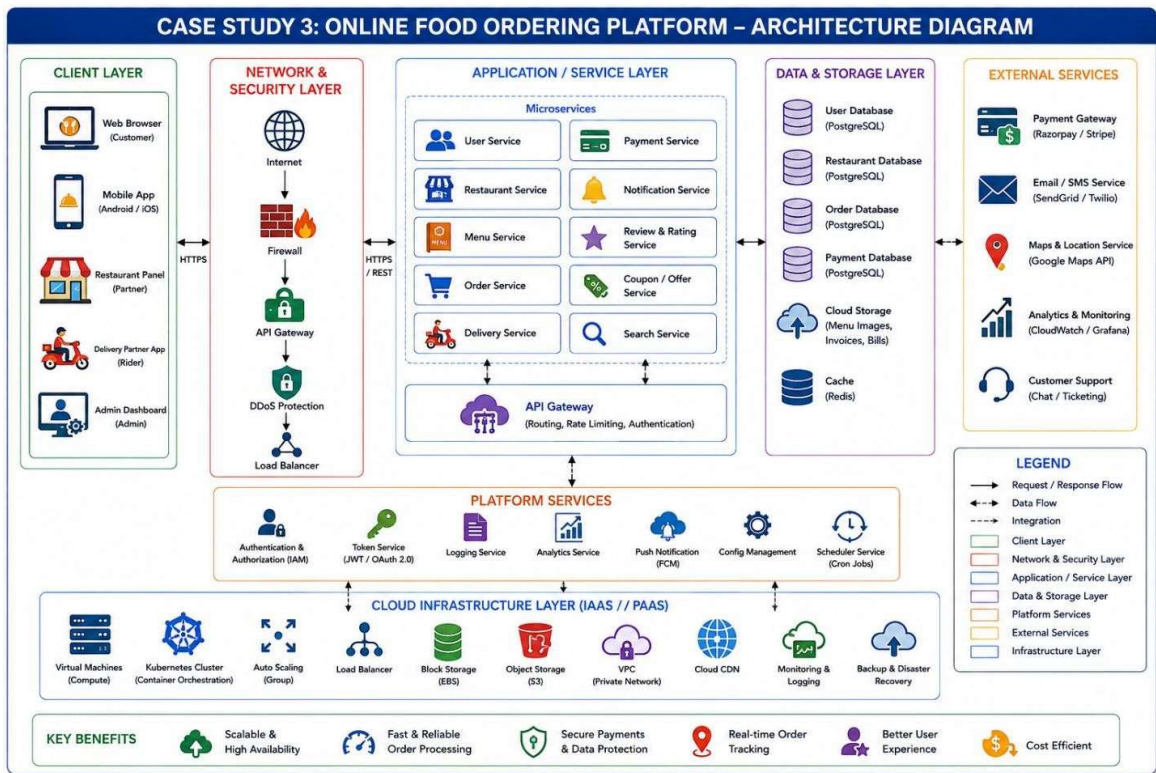
Security Features

- HTTPS
- JWT Authentication Tokens
- Role-Based Access Control (RBAC)
- Data encryption
- Secure payment processing

Benefits

- Scalable infrastructure
- High availability
- Fast order processing
- Secure customer data
- Real-time communication

Architecture Diagram



Conclusion

Cloud technology enables a reliable, scalable, and secure online food ordering platform capable of handling thousands of concurrent users efficiently.